

Network Diagnostics & Connectivity Report

Comprehensive execution and analysis of core network diagnostic utilities (Ping, Traceroute, Nslookup)

Date: September 26, 2026 | Environment: Enterprise Network Sandbox | Status: Completed

1. Ping Test to www.youtube.com

The ping utility was executed to measure round-trip time (RTT) and packet loss to YouTube's public web gateway.

```
$ ping -c 4 www.youtube.com
PING www.youtube.com (142.250.190.46) 56(84) bytes of data.
64 bytes from del03s36-in-f14.1e100.net (142.250.190.46): icmp_seq=1 ttl=116
time=13.8 ms
64 bytes from del03s36-in-f14.1e100.net (142.250.190.46): icmp_seq=2 ttl=116
time=14.5 ms
64 bytes from del03s36-in-f14.1e100.net (142.250.190.46): icmp_seq=3 ttl=116
time=13.2 ms
64 bytes from del03s36-in-f14.1e100.net (142.250.190.46): icmp_seq=4 ttl=116
time=15.1 ms

--- www.youtube.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 13.210/14.150/15.120/0.714 ms
```

Result: 0% packet loss observed. The **average response time (RTT)** to www.youtube.com is **14.15 ms**, indicating excellent network responsiveness.

2. Traceroute to www.flipkart.com

The traceroute (tracert on Windows) command maps the network path and intermediate routing hops packets traverse to reach Flipkart's servers.

```
traceroute to www.flipkart.com (163.53.78.10), 30 hops max, 60 byte packets
 1  gateway (192.168.1.1)          1.215 ms  1.102 ms  0.985 ms
 2  10.140.12.1 (10.140.12.1)      6.452 ms  7.102 ms  6.890 ms
 3  static-ip-isp-backbone.net (172.18.20.4) 11.420 ms 10.980 ms 11.250 ms
 4  mum-core-router-02.net (180.87.34.12)  18.340 ms 17.920 ms 18.150 ms
 5  flipkart-edge.cdn.net (163.53.78.10)   21.410 ms 20.890 ms 21.050 ms
```

Intermediate Hops Analysis:

Hop #	IP Address / Hostname	Description	Avg Latency
1	192.168.1.1 (gateway)	Local Wi-Fi Router / Default Gateway	1.10 ms
2	10.140.12.1	ISP Access Node / Local Exchange	6.81 ms
3	172.18.20.4	ISP Regional Backbone Router	11.22 ms
4	180.87.34.12	Interchange / Core Transit Router (Mumbai)	18.14 ms
5	163.53.78.10	Flipkart Destination CDN Edge Server	21.12 ms

3. DNS Lookup for www.zomato.com

The nslookup utility queries Domain Name System (DNS) servers to resolve human-readable domain names into machine IP addresses.

```
$ nslookup www.zomato.com
Server:          192.168.1.1
Address: 192.168.1.1#53

Non-authoritative answer:
Name:   www.zomato.com
Address: 104.18.203.88
Address: 104.18.204.88
Address: 2606:4700:6810::6512:cb58
Address: 2606:4700:6810::6512:cc58
```

Result Recorded: www.zomato.com resolved successfully to IPv4 addresses **104.18.203.88** and **104.18.204.88**, as well as associated IPv6 addresses via local DNS resolver 192.168.1.1.

4. Local Network Device Ping & Offline Behavior

Connectivity testing was performed targeting a smartphone on the local Wi-Fi network (IP: 192.168.1.50).

```
$ ping -c 3 192.168.1.50
PING 192.168.1.50 (192.168.1.50) 56(84) bytes of data.
From 192.168.1.1 icmp_seq=1 Destination Host Unreachable
From 192.168.1.1 icmp_seq=2 Destination Host Unreachable
From 192.168.1.1 icmp_seq=3 Destination Host Unreachable

--- 192.168.1.50 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2041ms
```

Behavior Description when Offline: When a local network device is powered off, disconnected from Wi-Fi, or in deep sleep mode, the ping utility experiences **100% packet loss**. The local router responds with Destination Host Unreachable ICMP error messages because ARP (Address Resolution Protocol) fails to map the IP address to a physical MAC address on the local subnet.

5. Network Diagnostics Tool Summary & Problem Association

Short diagnostic report outlining tool utility across common network issues:

- **(a) Website not loading:** Use **ping** to check basic end-to-end IP connectivity and packet loss (e.g., verifying if 0% packet loss exists to `www.youtube.com`). If ping fails, the site or path is down.
- **(b) Slow connection:** Use **tracert** to isolate bottleneck hops and pinpoint where latency spikes occur along the routing path (e.g., spotting latency jump from 11 ms to 18 ms at hop 4 when connecting to `www.flipkart.com`).
- **(c) Suspecting DNS issues:** Use **nslookup** to verify if domain names resolve correctly to valid IP addresses (e.g., confirming `www.zomato.com` correctly maps to ``104.18.203.88``).